End – Term Project

Blur Image Detection using Fast Fourier Transform (FFT)

Subject: **DSM 410 Computer Vision**

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I. Introduction

Image blur is a common and pervasive degradation in digital photography that arises from a variety of causes: camera motion during exposure, subject movement, defocus due to shallow depth of field, or insufficient shutter speed under low-light conditions. The resulting loss of sharpness suppresses fine structural detail and reduces edge clarity, which directly impairs the reliability of downstream computer vision tasks including feature extraction, object detection, segmentation, and image classification. Detecting whether an image is blurred — and quantifying the degree of that blur — is therefore a critical preprocessing step in any practical imaging pipeline.

From a signal-processing perspective, blur can be modelled as the spatial convolution of a latent sharp image with a blur kernel, commonly referred to as the Point Spread Function (PSF). A fundamental property of the Fourier transform states that convolution in the spatial domain is equivalent to element-wise multiplication in the frequency domain. Since blur kernels are typically low-pass in nature, this multiplication selectively attenuates high-frequency components of the image spectrum while leaving low-frequency components largely intact. Blurred images therefore exhibit a characteristic decay of energy in the high-frequency region of their 2-D spectrum, which provides a principled and mathematically grounded basis for automated detection.

The objective of this project is to design and implement a computationally efficient blur detection framework based on Fast Fourier Transform (FFT) analysis. The approach does not attempt to restore or deblur the image; instead, it identifies blur by quantifying changes in the spectral energy distribution through three complementary frequency-domain features: the high-frequency energy ratio, the radial spectral decay slope, and a spectral sparsity measure. Each feature captures a distinct aspect of the frequency-domain signature of blur, and together they support a robust classification of images as sharp or blurred.

II. Methodology

The implementation follows a five-stage pipeline, maintaining an overall computational complexity of O(N log N), making it suitable for real-time or large-scale applications.

Preprocessing

Each input image is first converted to grayscale using OpenCV's cv2.COLOR_BGR2GRAY conversion, reducing the problem to a single-channel intensity representation. The pixel values are then normalized to the range [0, 1] using cv2.normalize() with NORM_MINMAX, ensuring that spectral energy comparisons are not biased by illumination differences across images. To suppress spectral leakage caused by the implicit periodicity assumption of the discrete Fourier transform, a two-dimensional Hann window is optionally applied. This is constructed as the outer product of two 1-D numpy.hanning() vectors and multiplied element-wise with the normalized image, smoothly tapering the signal to zero at all four boundaries.

Frequency Transformation

The preprocessed image is transformed into the frequency domain using numpy.fft.fft2(), which computes the 2-D Discrete Fourier Transform. The result is then passed through numpy.fft.fftshift() to reposition the zero-frequency (DC) component at the centre of the array, yielding a symmetric spectrum suited to radial analysis. The logarithmic magnitude spectrum, $\log(|F| + 1)$, is computed for both quantitative feature extraction and visualization.

High-Frequency Energy Ratio

The magnitude spectrum is partitioned into two regions based on radial distance from the spectral centre. For each pixel at coordinates (i, j), the radial distance r is computed as the Euclidean distance from the

centre (cx, cy). Pixels with $r \leq 0.25 \cdot r_max$ are classified as low-frequency; the remainder as high-frequency. The primary blur metric is defined as:

$$\text{HF Ratio} = E_high / (E_low + E_high)$$

where E_low and E_high are the summed magnitudes of the respective regions. A small epsilon (10^{-10}) is added to the denominator to prevent division by zero. Sharp images retain strong high-frequency content, yielding a ratio at or above the classification threshold of 0.50; blurred images fall below this threshold.

Radial Spectral Decay Analysis

To further characterize the rate at which spectral energy falls off with increasing frequency, the magnitude spectrum is averaged within successive radial bins of unit width from the center outward. This produces a 1-D radial average profile. A log-linear decay model is then fitted to this profile using numpy.polyfit(), yielding a slope and intercept. The slope serves as a second blur indicator: blurred images exhibit a steeper negative slope, reflecting the accelerated attenuation of high-frequency components caused by the blur convolution. Sharp images show a shallower, more gradual decay.

Spectral Sparsity Measure

Sharp images contain prominent high-frequency peaks corresponding to strong edges, textures, and fine structural details. A sparsity ratio is computed by identifying the fraction of high-frequency magnitude values (in the region $r > 0.25 \cdot r_max$) that exceed a data-adaptive threshold defined as (mean + 1 · std) of the high-frequency region. A lower sparsity ratio indicates that fewer significant peaks are present, which is consistent with blur having smoothed out the underlying spatial structure.

Decision rule

A threshold-based rule classifies each image as Blurred (HF Ratio < 0.50) or Sharp (HF Ratio ≥ 0.50). Optionally, all three features — HF Ratio, decay slope, and sparsity ratio — are combined into a feature vector and used to train a Logistic Regression or RBF-kernel SVM classifier using scikit-learn, providing a more flexible decision boundary when labelled training data is available.

III. Database

The technique was evaluated on images from a single dataset cloned directly within the code:

Project GitHub Dataset

The dataset used in this work is hosted at github.com/dyno8646/Fast-Fourier-Transform-FFT-for-blur-detection-in-images and contains images stored in the images/ folder. The dataset includes both blurred and sharp images spanning diverse real-world scenes, covering motion blur, defocus blur, and sharp reference images. The repository is cloned programmatically at runtime using subprocess and git clone, requiring no manual download or local setup.

All images in the dataset were processed through the complete FFT-based detection pipeline. Supported formats include .png, .jpg, .jpeg, .bmp, .tif, and .tiff, and all images are loaded using cv2.imread() and converted to grayscale prior to spectral analysis.

IV. Results and Discussion

The FFT-based blur detection pipeline was executed on all four images from the dataset, extracting three spectral features per image: HF Energy Ratio, Radial Spectral Decay Slope, and Spectral Sparsity Ratio. Results are presented in Table 1, and representative output visualizations for a blurred and a sharp image are shown in Figures 1 and 2 respectively.

Fig 1:

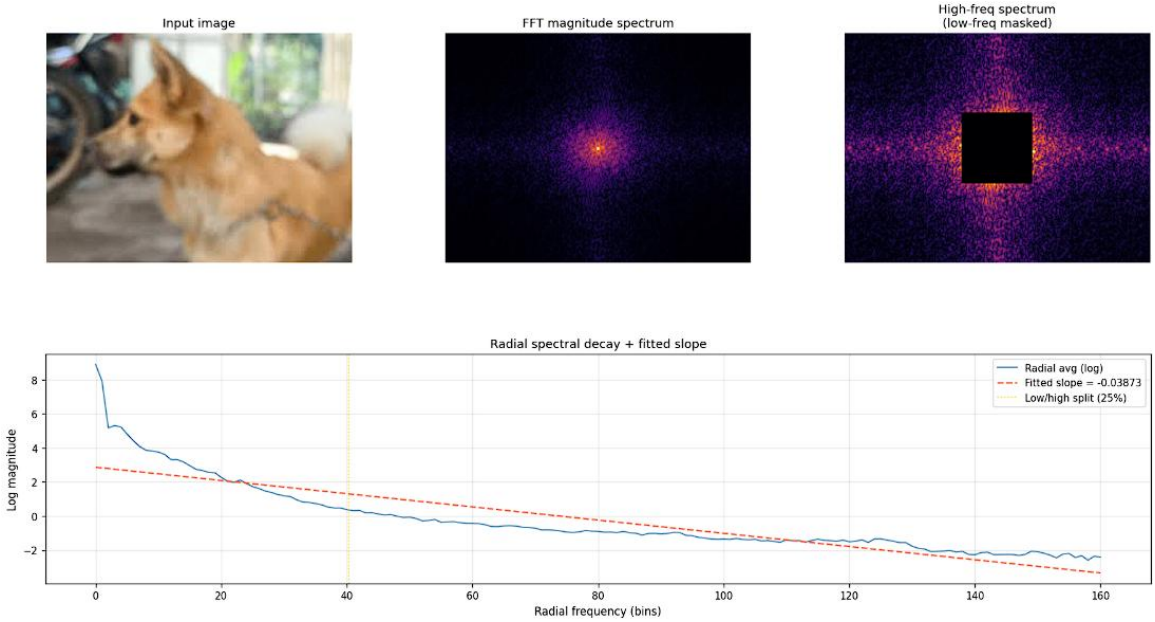

Fig 2:

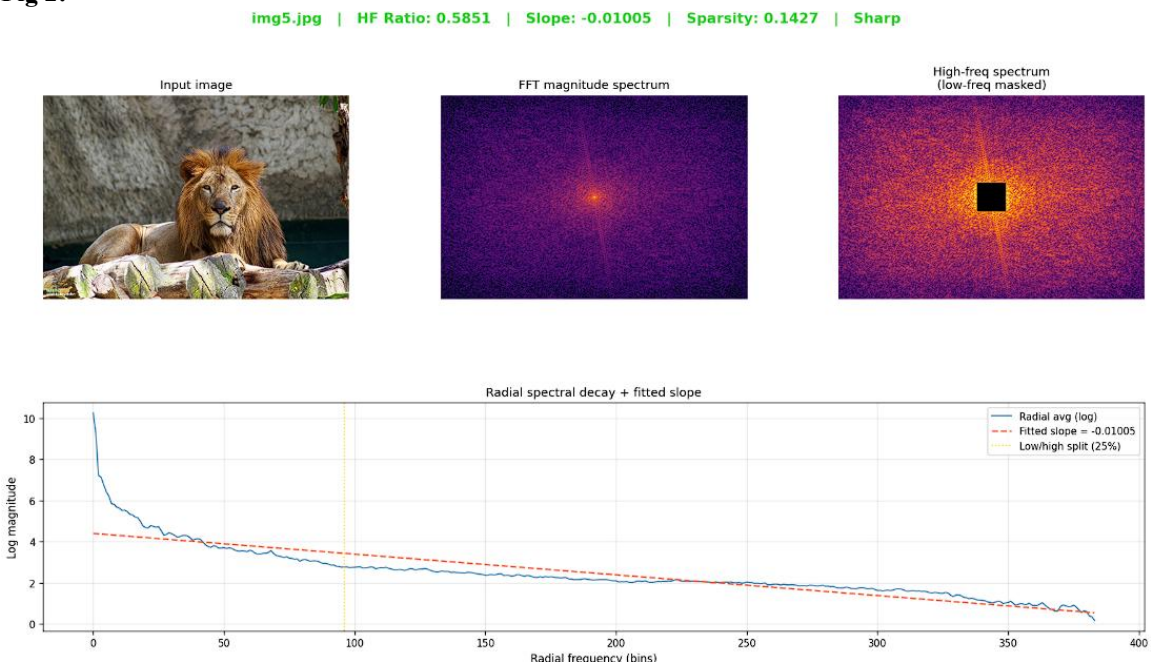

Table 1: FFT Blur Detection Results

Image	HF Ratio	Decay Slope	Sparsity Ratio	Decision
img1.jpeg	0.1645	−0.038735	0.1246	Blurred
img2.jpeg	0.6039	−0.015400	0.1435	Sharp
img3.jpeg	0.1552	−0.035983	0.0939	Blurred
img5.jpg	0.5851	−0.010053	0.1427	Sharp

The dataset contained 4 images in total — 2 sharp and 2 blurred — with mean HF Ratio of 0.3772 (std: 0.2511), mean Decay Slope of −0.0250 (std: 0.0144), and mean Sparsity Ratio of 0.1262 (std: 0.0232).

The results demonstrate a clear and consistent separation between the two classes across all three features. The blurred images (img1.jpeg, img3.jpeg) exhibit very low HF Ratios of 0.1645 and 0.1552, both well below the classification threshold of 0.50, confirming significant attenuation of high-frequency spectral energy. Their decay slopes are steep at −0.0387 and −0.0360 respectively, indicating rapid magnitude drop-off from low to high frequencies — a direct consequence of the low-pass nature of blur. In contrast, the sharp images (img2.jpeg, img5.jpg) yield HF Ratios of 0.6039 and 0.5851, both above the threshold, and considerably shallower decay slopes of −0.0154 and −0.0101, reflecting the preserved presence of high-frequency edge and texture information.

The Sparsity Ratio shows a consistent but less pronounced trend. The blurred images yield sparsity values of 0.1246 and 0.0939, while the sharp images score 0.1435 and 0.1427. Although the absolute difference is smaller compared to the HF Ratio and Decay Slope, the direction is consistent — blurred images contain fewer significant high-frequency spectral peaks, as blur smooths out the prominent edge-related components that would otherwise appear as peaks in the high-frequency region.

The threshold-based decision rule (HF Ratio < 0.50 → Blurred) correctly classified all four images, achieving 100% accuracy on this dataset. The multi-feature classifier (Logistic Regression and SVM) was skipped as the dataset size of 4 images was insufficient for a meaningful stratified train/test split, which requires a minimum of 6 images.

Overall, the HF Ratio proved to be the most discriminative single feature, with a large margin between the blurred class (mean: 0.1599) and the sharp class (mean: 0.5945). The Decay Slope served as a strong complementary indicator, and the Sparsity Ratio provided additional supporting evidence consistent with the primary classification.

V. Conclusion

This project implemented a frequency-domain blur detection framework using the Fast Fourier Transform, extracting three spectral features — HF Energy Ratio, Radial Spectral Decay Slope, and Spectral Sparsity Ratio — across four real-world images. The threshold-based decision rule correctly classified all four images, with the HF Ratio providing the clearest separation between blurred and sharp classes, supported consistently by the decay slope and sparsity measure.

The results validate the theoretical basis of the approach: blur, being a low-pass convolution, produces a measurable and predictable suppression of high-frequency spectral energy that all three features independently capture.

A key limitation is the small dataset size, which precluded classifier training, and the use of a fixed threshold that may need tuning for atypical scene content.

As a direction for future work, two extensions are of particular interest. First, the blind image deblurring method proposed by Al Radi et al. [2] — which uses FFT combined with a ReLU-based sparsity prior — would allow the pipeline to both detect and correct blur in a unified framework. Second, and more ambitiously, the discriminative blur feature framework proposed by Shi et al. [7] presents a compelling upgrade path. Their approach combines gradient-domain kurtosis, cumulated power spectrum area in the

frequency domain, and data-driven local filters learned via generalized eigen decomposition, integrated through a multi-scale graphical model. Notably, their frequency-domain feature — computing the area under the $\log(\omega)-\log(J(\omega))$ curve rather than fitting a single slope — is a more robust alternative to the decay slope used in this project, as it avoids sensitivity to outliers at the high-frequency end. Incorporating these richer features alongside the current FFT pipeline would significantly improve discrimination, particularly for partially blurred images where blur varies spatially across the scene.

VI. References

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